BCS ICT FEST 2026 · MODULE 12 — PROJECT SHOWCASE

MIRRORWORLD

A constraint-aware campus digital twin for safer scheduling decisions

Test the future of your campus before you change it.

ABSTRACT

Universities must adjust timetables when rooms close, enrolment changes or staff become unavailable. MIRRORWORLD provides a digital model of campus rooms, instructors, student groups and scheduled sessions. Users can create a what-if scenario, inspect affected resources, generate alternative schedules with the OR-Tools CP-SAT solver, and compare plans against the baseline. The interface reports constraint issues and operational indicators—including room use, workload, movement and estimated energy—so decision-makers can review trade-offs before applying a change. The current demonstration uses a clearly labelled synthetic campus dataset.

THE PROBLEM / THE RESPONSE

THE PROBLEM	THE RESPONSE
A timetable change can affect room capacity, laboratory requirements, instructor availability and student-group clashes at once. Manual trial-and-error makes those side effects hard to see before a decision is made.	MIRRORWORLD keeps a baseline, applies changes in a separate scenario, traces related resources and proposes feasible alternatives. It shows the measured differences and remaining issues for review.

ONE DECISION WORKFLOW

1 MODEL	2 DETECT	3 SIMULATE	4 OPTIMIZE	5 COMPARE

IMPLEMENTATION

The web interface is built with React and TypeScript; a Python/FastAPI service provides the application API, with SQLAlchemy and a local SQLite database. Google OR-Tools CP-SAT searches for schedules while enforcing room, instructor and group non-overlap, room capacity, facility fit, availability and operating-window rules. Scenario metrics and dependency views support inspection and comparison. The solver reports its status; an infeasible scenario is not presented as a valid plan. Core technologies include React, FastAPI, SQLAlchemy, SQLite and OR-Tools.

DEMO DATASET · 4 buildings · 24 rooms · 26 instructors · 24 cohorts · 112 weekly sessions · 1,128 students

EVIDENCE, IMPACT & DEMONSTRATION

What the current prototype can show

The figures below are read from the running application's seeded Fall 2026, Week 7 dataset. They describe the synthetic baseline only; they are not a claim of post-optimization improvement or results from a real university.

BASELINE INDICATOR	CURRENT VALUE	INTERPRETATION
Schedule	112 / 112 sessions assigned	Synthetic weekly timetable across 24 rooms.
Conflicts	3 total	1 room, 1 instructor and 1 cohort collision in the seeded baseline.
Hard-constraint issues	6 flagged	Includes a 20-seat capacity deficit, a facility mismatch and an availability issue, alongside the clashes.
Room use	14.6% weekly; 79.2% peak slot	184.5 occupied room-hours out of 1,260 available.
Energy model	1,406.2 kWh / week	Estimated from room power ratings and scheduled hours; not meter data.

WHAT IS DISTINCTIVE

Digital twins and timetable optimization are established areas. MIRRORWORLD's contribution is their integration into one campus decision workflow: change a condition, identify connected resources, solve under explicit hard constraints, and compare alternatives using transparent indicators. This project does not claim to invent those underlying methods; it demonstrates a usable combination for disruption-aware university planning.

SUSTAINABLE DEVELOPMENT GOALS

SDG 9	Evaluate how existing rooms and laboratories can be used more effectively through constraint-aware planning.
SDG 11	Support more resilient campus operations by testing disruption responses before changing a schedule.
SDG 12	Compare modelled energy use across plans to inform resource-conscious decisions; energy values remain estimates.

LIMITATIONS & NEXT VALIDATION

The current campus is synthetic, so its metrics do not establish impact at a real institution. Energy is an estimate based on modelled room power and scheduled hours; movement uses a simplified building-distance model. The next validation step is to test an anonymized institutional timetable, verify scenario results with campus planners, and report only measured before/after outcomes. All application logic and results should be distinguished from the credited open-source libraries used to build the prototype.

LIVE DEMONSTRATION OUTLINE

01 BASELINE	Show the campus model and its current conflict/resource indicators.
02 DISRUPT	Create a room-closure scenario and inspect affected resources in the dependency view.
03 RECOVER	Run the optimizer; show solver status and the resulting candidate schedule.
04 EXPLAIN	Compare the plan with baseline, explain the trade-offs and state what remains unresolved.

ORIGINALITY & SAFETY · Software-only demonstration; no hazardous hardware. The prototype uses a generated synthetic dataset and open-source software dependencies. Team eligibility, student IDs and originality declaration must be confirmed by the team before submission.